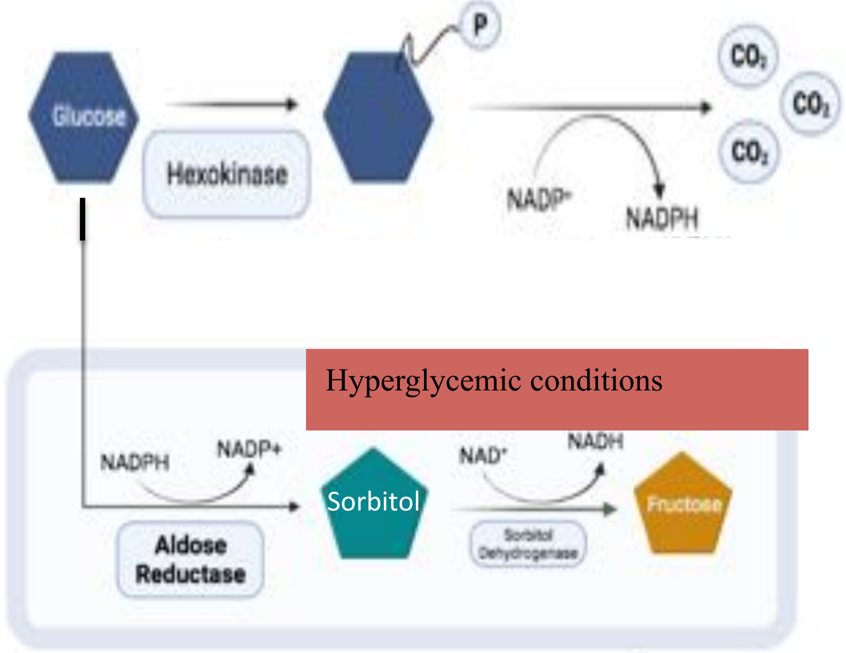
Aldose Reductase and Diabetic Retinopathy

Glycolysis

Polyol Pathway



- AR is a well-studied drug target
- It is structurally well- characterized
- Equilibrium unfolding of AR was not studied.
- Is folding of AR relevant to its function?

Abnormality	**Compound**	**Aim of treatment**	**Status of RCT**
Polyol pathway↑	Aldose reductase inhibitors	**Nerve sorbitol ↓**	
	Sorbinil		Withdrawn (AE)
	Tolrestat		Withdrawn (AE)
	Ponalrestat		Ineffective
	Zopolrestat		Withdrawn (marginal effects)
	Zenarestat		Withdrawn (AE)
	Lidorestat		Withdrawn (AE)
	Fidarestat		Effective in RCTs, trials ongoing
	AS-3201		Effective in RCTs, trials ongoing
	Epalrestat		Marketed in Japan

Diabetes Care Apr 2005, 28 (4) 956-962

Unfolding Studies of AR

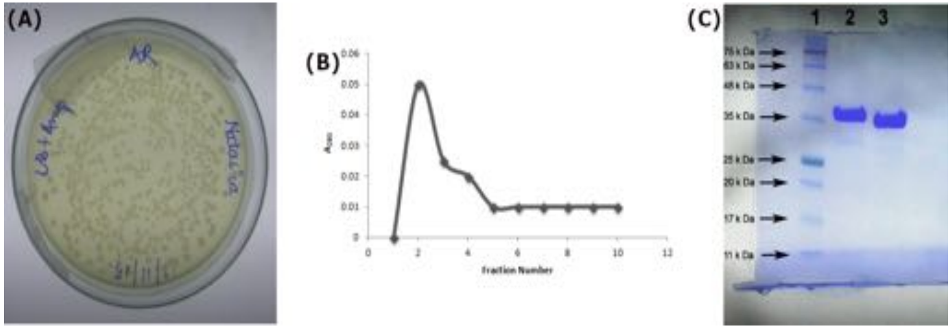
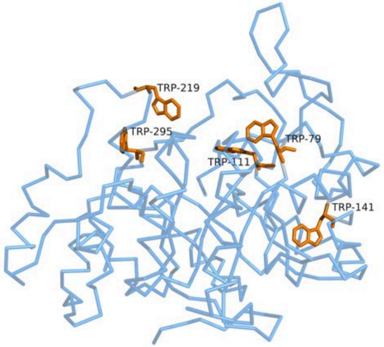


Figure. (A) Overexpression (B) Elution Profile (c) His-tagged (lane 2) and Cleaved AR (lane 3).



Tryptophan Fluorescence:

$\lambda_{excitation} = 295\ nm,\ \lambda_{emission} = 300 - 400\ nm$

ANS Fluorescence:

$\lambda_{excitation} = 370\ nm,\ \lambda_{emission} = 400 - 600\ nm$

Figure. Tryptophan residues in AR structure (PDB ID: 1ADS).

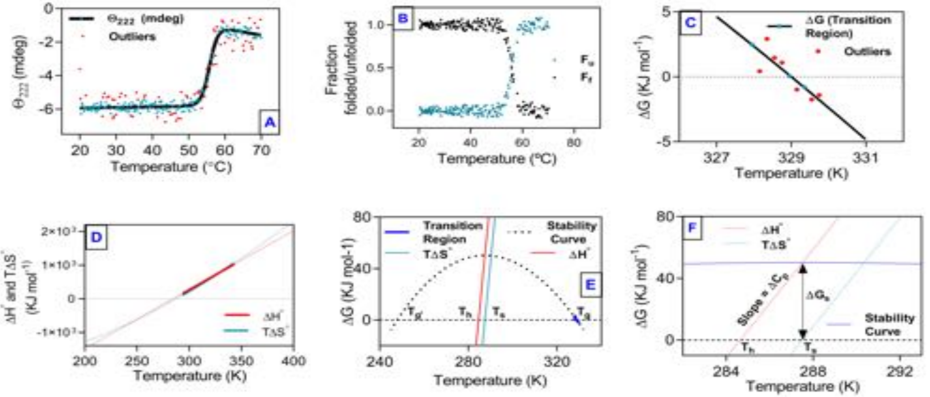


Figure. Thermal denaturation studies.

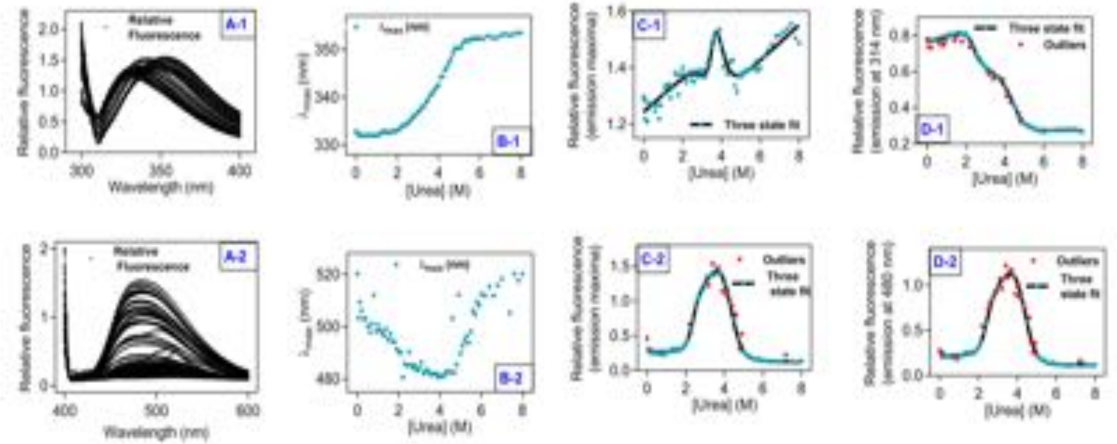


Figure. Fluorescence spectroscopic study of urea-induced equilibrium unfolding.

Unfolding Studies of AR

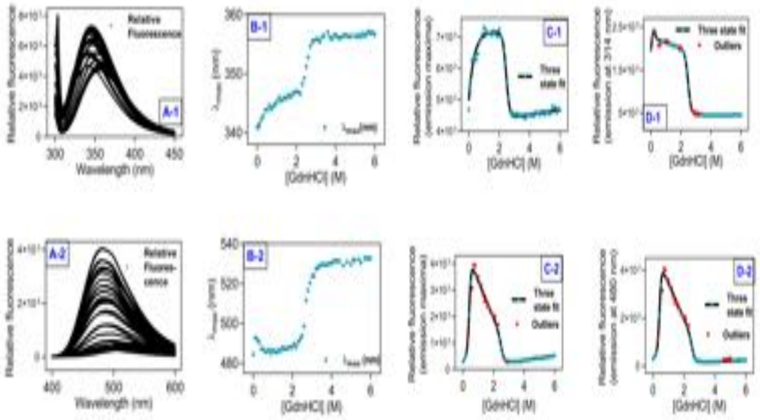


Figure. Fluorescence spectroscopic study of GdHCL-induced equilibrium unfolding.

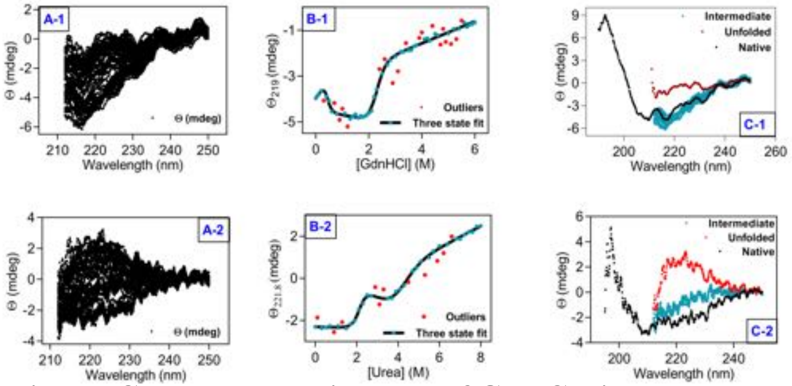


Figure. CD spectroscopic study of GdHCL-induced equilibrium unfolding.

Chemical unfolding	Probe	$\Delta G_{(N-I)}$ (KJ mol^{-1})	$\Delta G_{(I-U)}$ (KJ mol^{-1})	$m_{(N-I)}$ (KJ mol^{-1} M^{-1})	$m_{(I-U)}$ (KJ mol^{-1} M^{-1})	$C_{m(N-I)}$ (M)	$C_{m(I-U)}$ (M)		
	GdnHCl [F_{314}]	9.47 ± 1.69	58.34 ± 1.14	48.34 ± 7.69	23.56 ± 0.46	0.16	2.48		
	GdnHCl-ANS [F_{480}]	16.48 ± 0.54	57.26 ± 3.33	38.99 ± 1.15	22.38 ± 1.29	0.42	2.56		
	Urea [F_{314}]	30.88 ± 4.47	35.04 ± 4.16	13.11 ± 2.01	7.99 ± 0.84	2.355	4.49		
	Urea-ANS [F_{480}]	28.61 ± 2.39	38.6 ± 2.14	12.1 ± 1.13	8.95 ± 0.46	2.36	4.32		
	GdnHCl [Θ_{219}]	13.68 ± 5.78	34.13 ± 3.75	32.17 ± 11.63	14.9 ± 1.58	0.43	2.29		
	Urea [$\Theta_{221.8}$]	30.53 ± 6.16	24.43 ± 6.25	14.12 ± 3.03	5.99 ± 1.30	2.16	4.08		
Thermal unfolding	**Probe**	**ΔG_u (KJ mol^{-1})**	**T_g (K)**	**ΔC_p (KJ mol^{-1} K^{-1})**	**ΔH_g (KJ mol^{-1})**	**ΔS_m (KJ mol^{-1})**	**$T_{g`}$ (K)**	**T_h (K)**	**T_s (K)**
	Temperature [Θ_{222}]	50.25	329 ± 0.01	17.57	779.20 ± 15.36	2.37	248	284.6	287.5

Table. Summary of equilibrium unfolding studies.

Thermal Denaturation: Two State

$$Y = \frac{(A_n + b_N * T) + (A_u + b_U * T) * exp\left(\left(\frac{\Delta H_g}{R}\right) * \left(\frac{1}{Tg} - \frac{1}{T}\right)\right)}{1 + \exp\left(\left(\frac{\Delta H_g}{R}\right) * \left(\frac{1}{Tg} - \frac{1}{T}\right)\right)} \quad (1)$$

$$F_u = \frac{Y_n - Y}{Y_n - Y_u} \quad (2)$$

$$K = \frac{F_u}{1 - F_u} \quad (3)$$

$$\Delta G = -RT \ln k \quad (4)$$

$$\Delta H_T = \Delta H_g + \Delta C_p (T - T_g) \quad (5)$$

$$\Delta S_T = \frac{\Delta H_g}{T_g} + \Delta C_p * LN\left(\frac{T}{T_g}\right) \quad (6)$$

$$\Delta G_u = \Delta H_g * \left(1 - \frac{T}{T_g}\right) + \Delta C_p * \left(T - T_g - \left(T * LN\left(\frac{T}{T_g}\right)\right)\right) \quad (7)$$

$$\Delta C_p = -251 + 0.19 * [\Delta ASA] \quad (8)$$

A_n *and* A_u	Native and unfolded state baseline intercepts, respectively
b_N *and* b_U	Native and unfolded baseline slopes, respectively.
ΔH_T *and* ΔS_T	Enthalpy change and entropy change at temperature T
T_h, T_s, *and* T_g	The temperatures at which ΔH, ΔS, and ΔG are zero
ΔG_s	Stabilization free energy relative to the unfolded state
T	Absolute Temperature

T_g	Melting point
Δh_m	Enthalpy change at T_g
F_u	Fraction of folded protein
K	Equilibrium Constant
ΔG	Free energy Change
R	Universal Gas Constant

Protein Science 1995, 4 (10), 2138-2148; Protein stability curves. Biopolymers 1987, 26 (11), 1859-1877

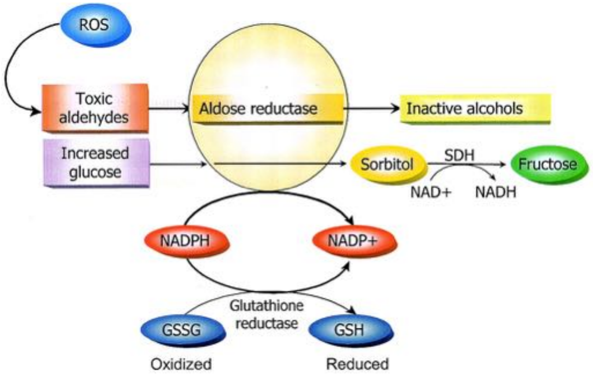
Chemical-Induced Unfolding: Three state

$$Y = \frac{(A_n + b_N * [D]) + (A_i + b_I * [D]) * \exp\left(-\left(\frac{\Delta G_{(N-I)}}{R * T} - \frac{m_{(N-I)} * [D]}{R * T}\right)\right) + (A_u + b_U * [D]) * \exp\left(-\left(\frac{\Delta G_{(N-I)}}{R} * T - \frac{m_{(N-I)} * [D]}{R * T}\right)\right) * \exp\left(-\left(\frac{\Delta G_{(I-U)}}{R * T} - \frac{[m(\quad]_{I-U}) * [D]}{R * T}\right)\right)}{1 + \exp\left(-\frac{\Delta G_{(N-I)}}{R * T} - \frac{m_{(N-I)} * [D]}{R * T}\right) - \exp\left(-\left(\frac{\Delta G_{(N-I)}}{R * T} - \frac{(m_{(N-I)} * [D])}{R * T}\right)\right) * \exp\left(-\left(\frac{\Delta G_{(I-U)}}{R * T} - \frac{m_{(I-U)} * [D]}{R * T}\right)\right)} \quad (9)$$

A_n, A_u, *and* A_i	The native, unfolded, and intermediate baseline intercepts, respectively
b_N, b_U, *and* b_I	The native, unfolded, and intermediate baseline slopes, respectively
[D]	Denaturant concentration in a molar unit.
$m_{(N-I)}$ *and* $m_{(I-U)}$	Denaturant gradient for native to intermediate and intermediate to unfolded state, respectively.
$\Delta G_{(N-I)}$ *and* $\Delta G_{(I-U)}$	Stabilization free energy of intermediate state relative to the native and unfolded state

Curr Protoc Protein Sci. 2013;Chapter 28:Unit28.4

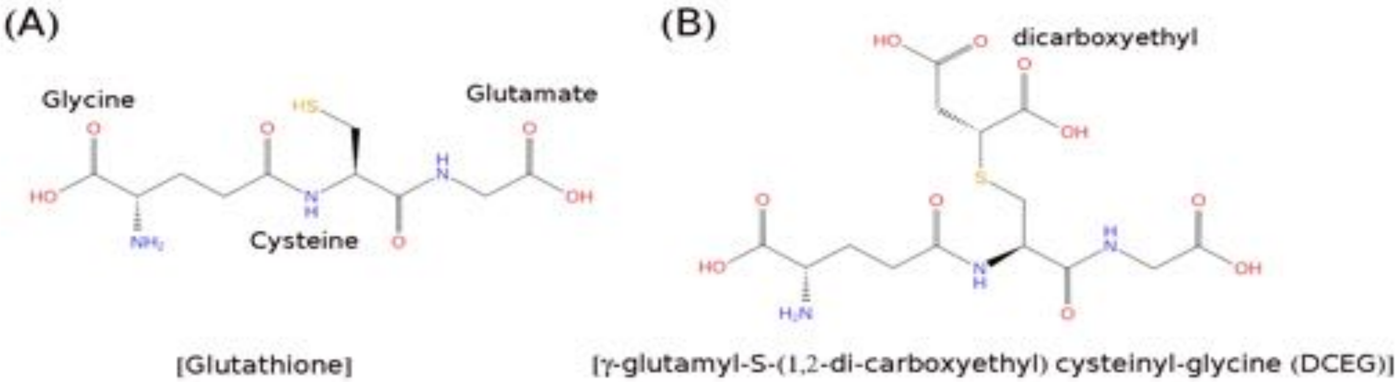
Aldose Reductase and Glutathione Metabolism



Diabetes 2005, 54 (6), 1615–1625

Inhibitor	K_i	Inhibition pattern
	mM	
Glu	No inhibition	
Gly	No inhibition	
Cys	4.018 ± 0.094	UC
γGlu-Cys	0.771 ± 0.038	UC
Cys-Gly	0.258 ± 0.020	UC
γGlu-Cys-Gly (GSH)	1.523 ± 0.094	UC
S-Lactoylglutathione	0.680 ± 0.121	UC
S-Propylglutathione	0.589 ± 0.120	UC
S-Hexylglutathione	0.784 ± 0.137	UC
Glutathionesulfonic acid	1.614 ± 0.446	UC
N-Ethylsuccinimidyl-GS	5.133 ± 0.829	UC
GS-menadione	0.001 ± 0.0001	UC
S-(1,2-Dicarboxyethyl)glutathione	0.197 ± 0.019	C
S-(*p*-Nitrobenzyl)glutathione	0.212 ± 0.048	C

Journal of Biological Chemistry 2000, 275 (28), 21587-21595



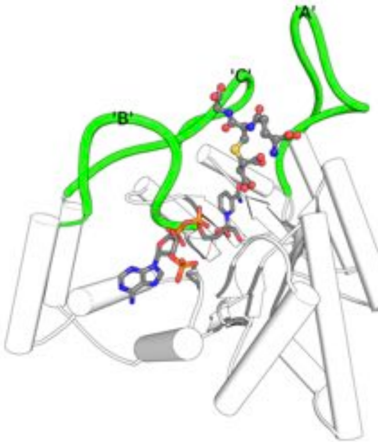
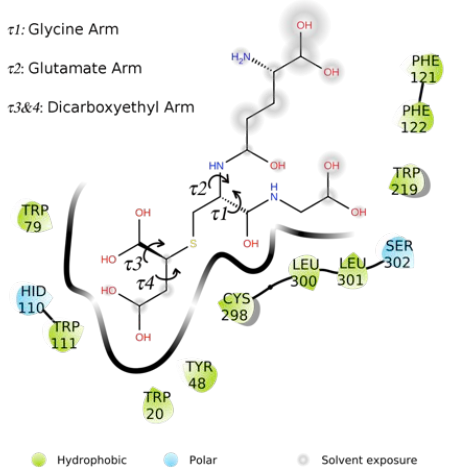
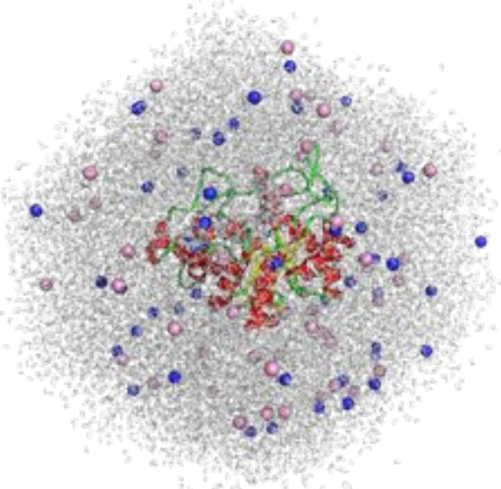


Figure. Crystal structure of AR:NADPH:DCEG (PDB ID: 2F2K)

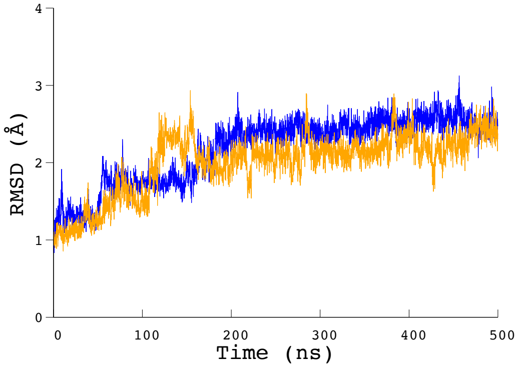


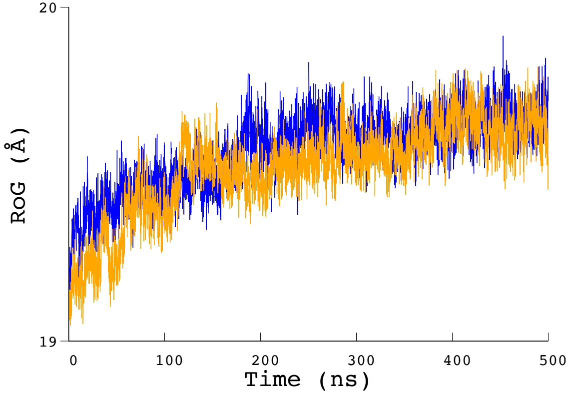
Can dynamics of DCEG in AR binding pocket reveal its specificity determinants ?

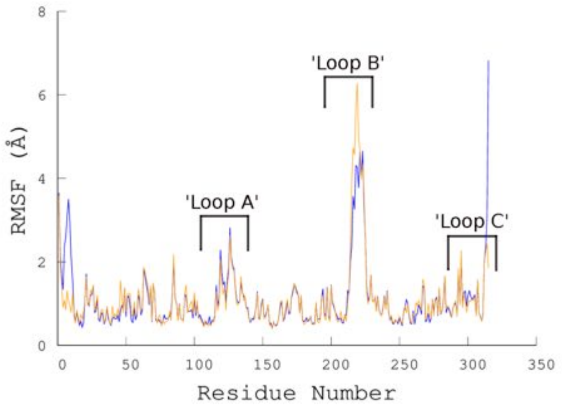
The MD simulation system and trajectory parameters

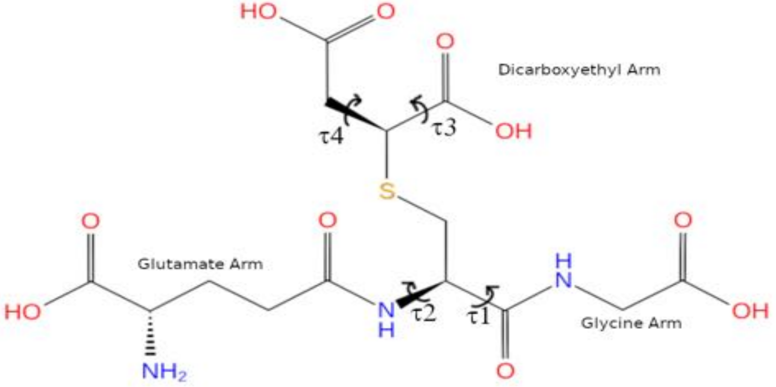


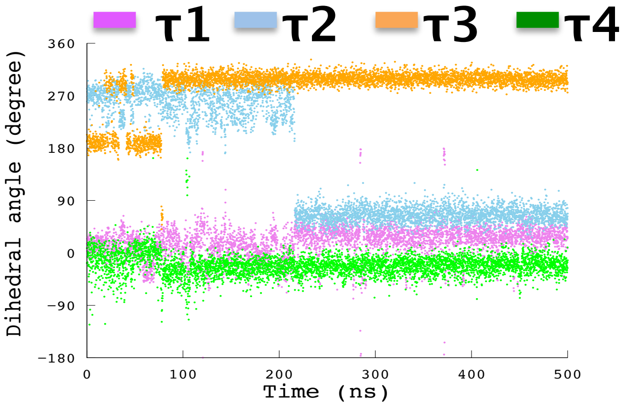
Simulation System:
Aldose Reductase
NADPH
DCEG
Chloride ions: 52
Sodium ions: 53
Water: 19270



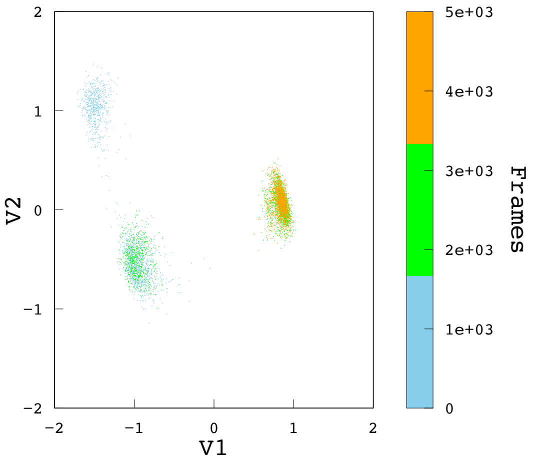


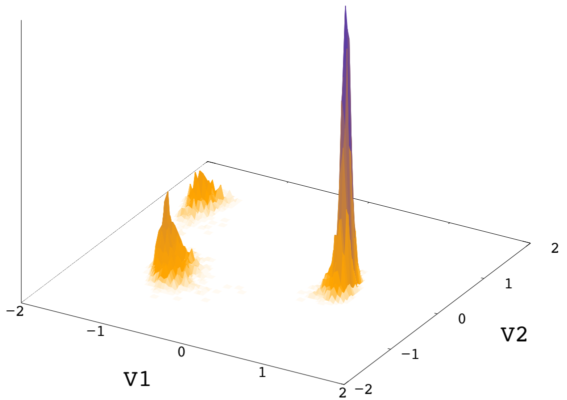


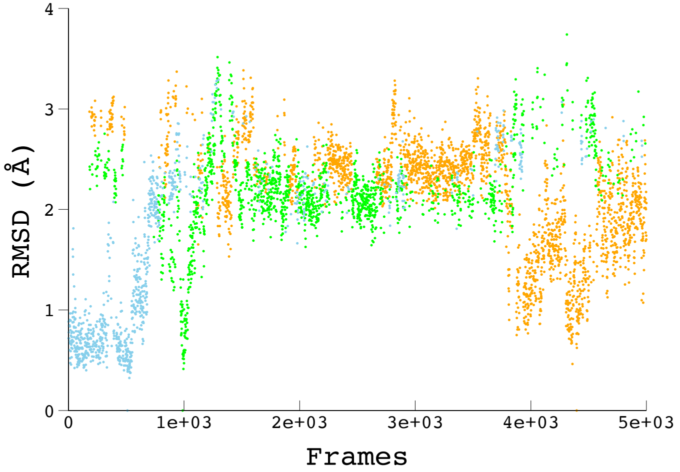


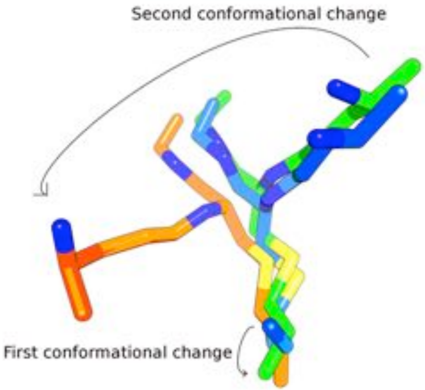


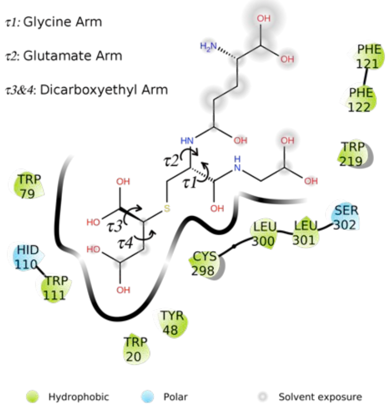
Conformational dynamics of DCEG











Major conformational changes:

1. Slight rearrangement in anion binding pocket
2. Glutamate arm of DCEG moves close to loop C (specificity determining region)

Movement of Cys-298 Regulates Glutatione Binding

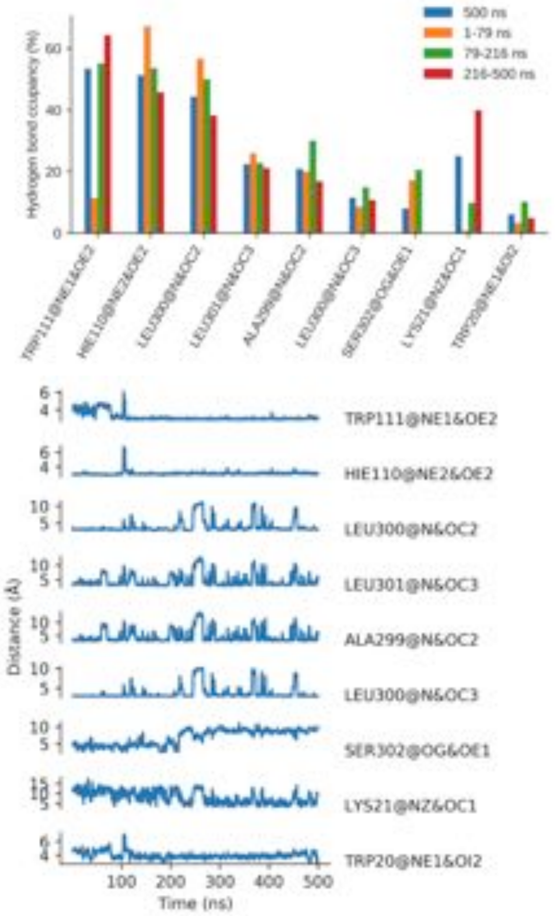
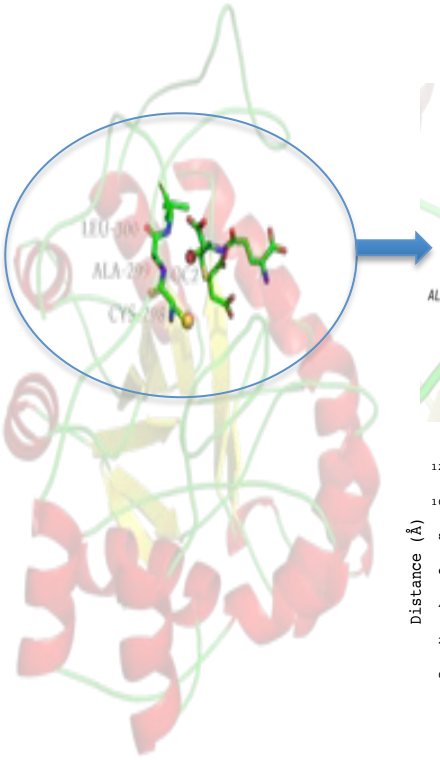
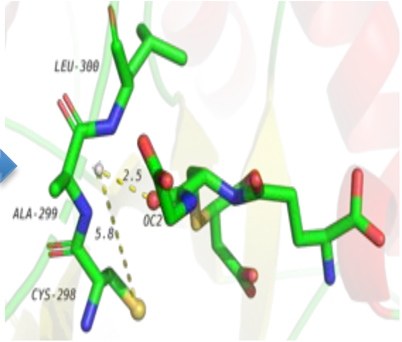
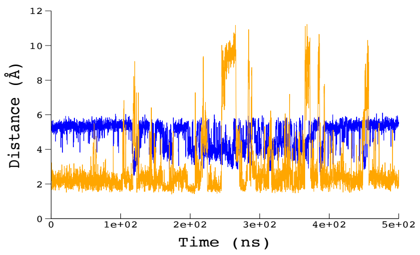


Figure. Hydrogen bond analysis.







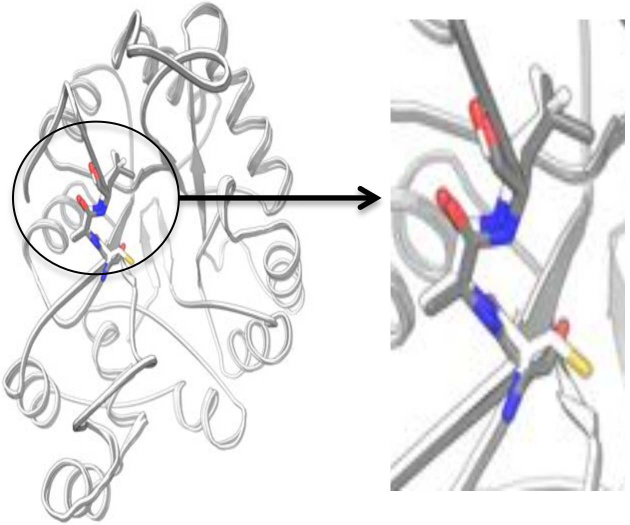


Figure S7. Structural superposition of wild type AR (White) and Cys298Ser mutant of AR (Grey) show relative position of side-chain Sulphur of Cys-298 and side-chain Oxygen of Ser-298 (PDBID: 3Q65 and 3Q67, respectively from *J. Biol. Chem. 2011, 286 (8), 6336–6344*).

Dynamics of DCEG into the Binding Pocket of AR

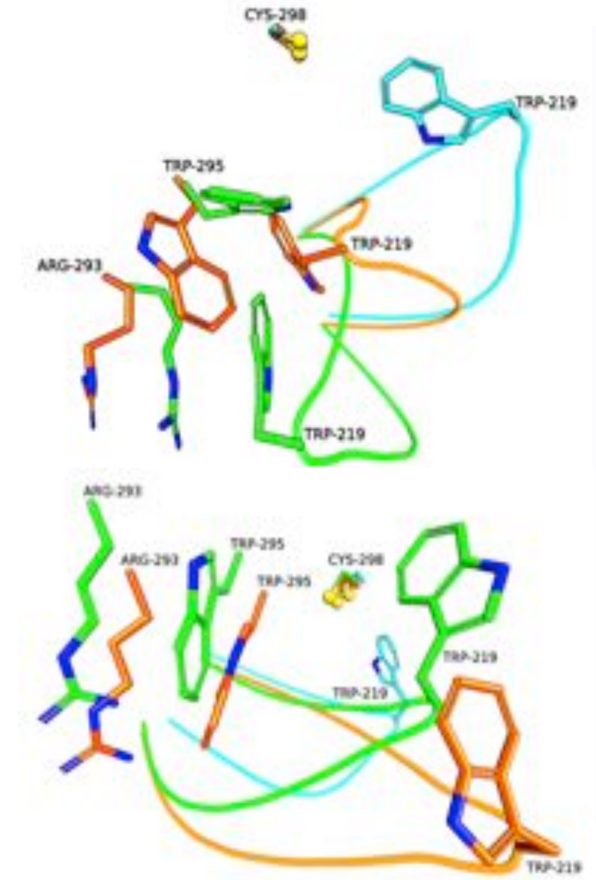
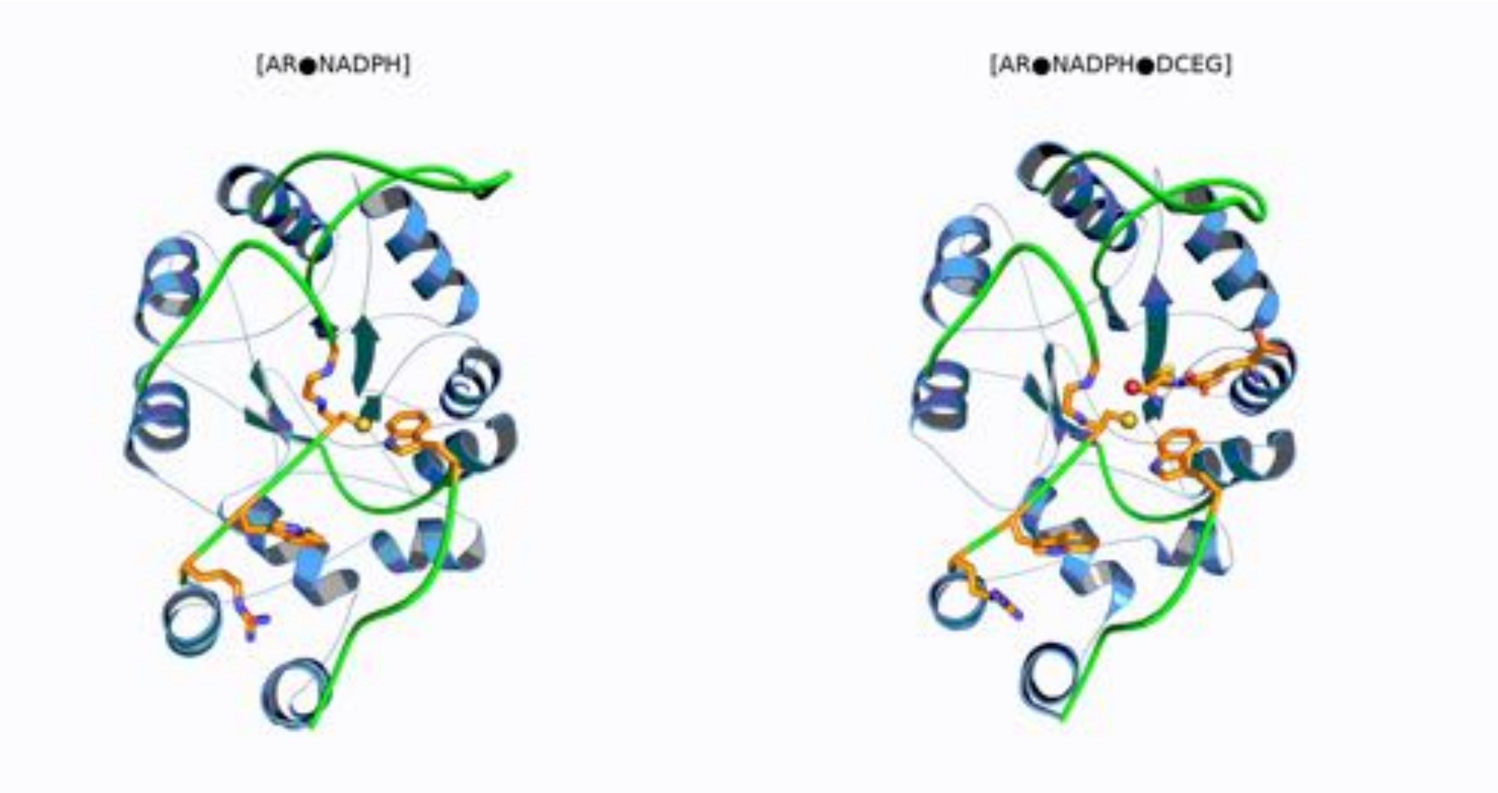


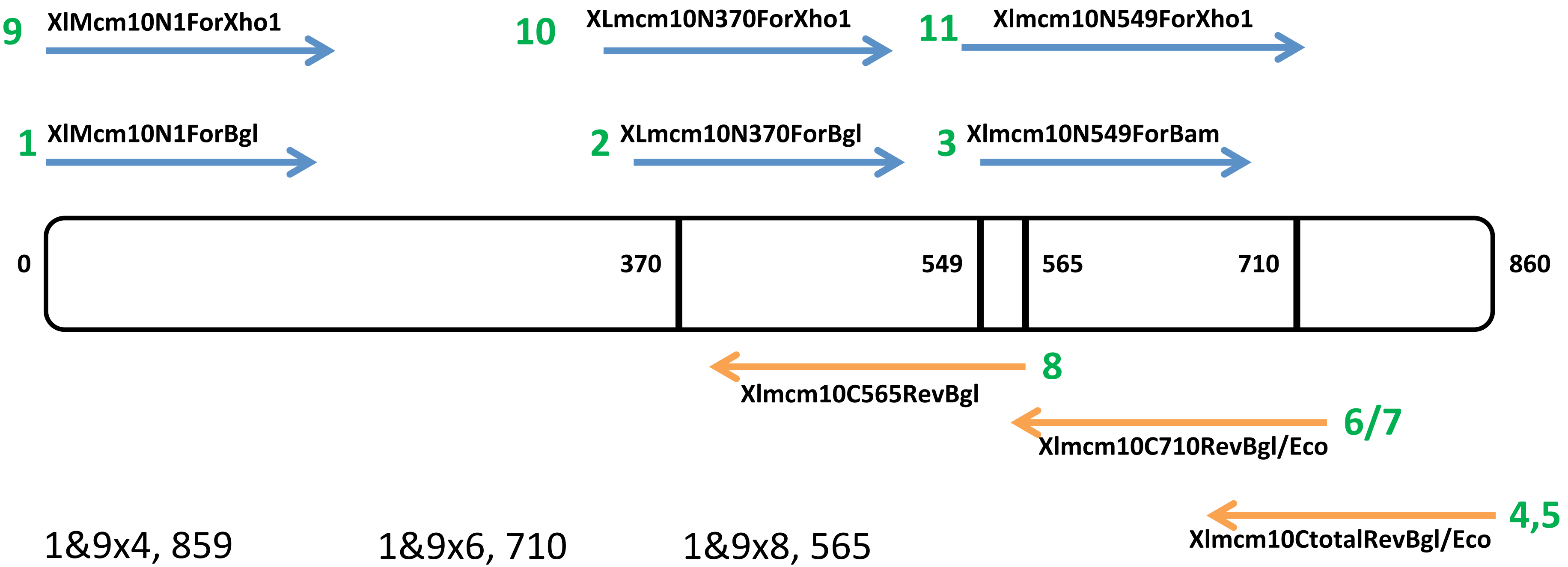
Figure. Opening of the Binding Pocket (Cyan: Closed; Orange: Intermediate; Green: Opened)



- The mechanism of recognition of DCEG by AR involves loop-C residues (Cys-298, Lla-299, and Leu-300. Cys-298) where Cys-298 functions as a gate to access the backbone nitrogen of loop-C residues.

Journal of Biomolecular Structure and Dynamics 2022, 40 (15), 6880–6888.

Does xMCM10 have Primase domain?

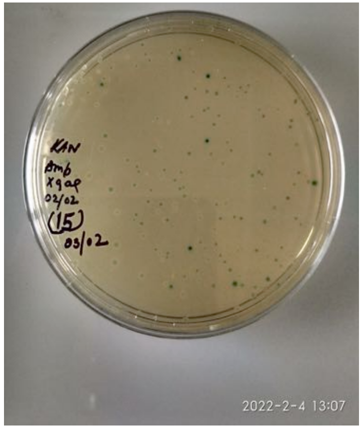


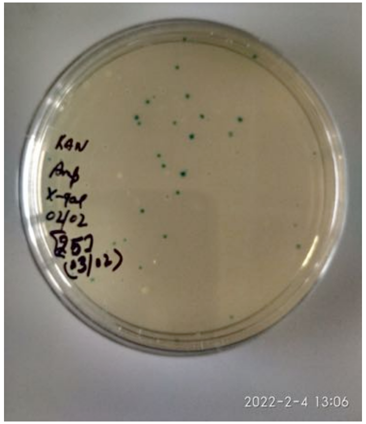
1&9x4, 859
2&10x4, 491
3&11x4&5, 312

1&9x6, 710
2&10x6, 341
3&11x6&7, 162

1&9x8, 565
2&10x8, 196
3&11x8, 17??

Cloning of xMCM10 Subdomains





Genome-wide Association Study of Carbapenem resistant *Acinetobacter baumannii*

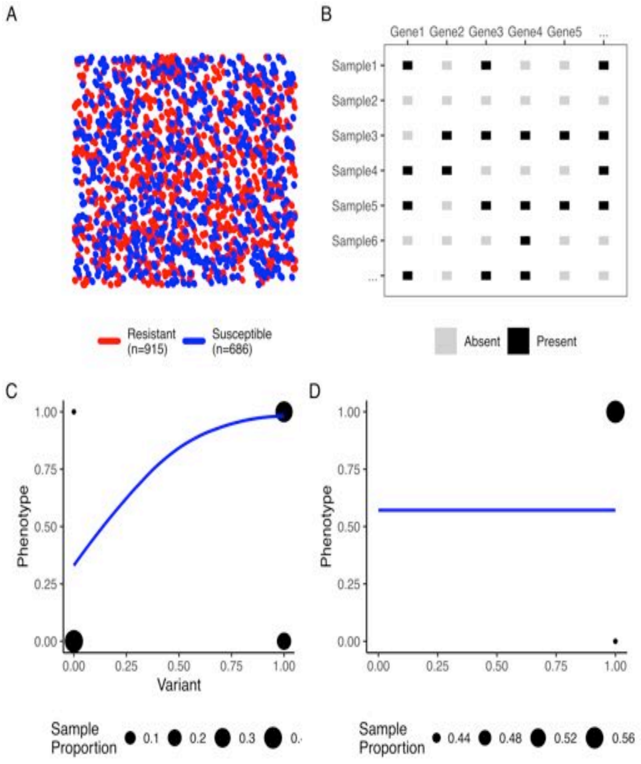


Figure. An introduction to GWAS

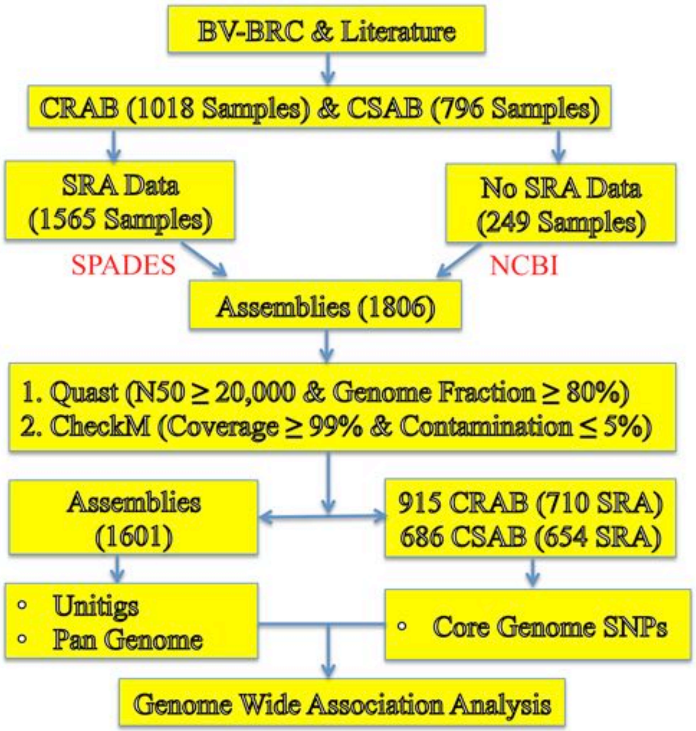


Figure. Overview of Study design

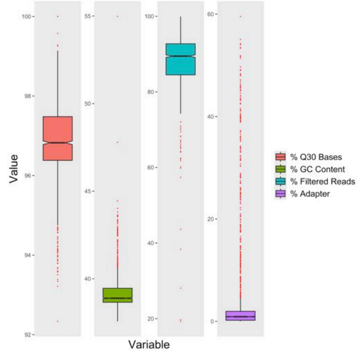


Figure. Quality of SRA data.

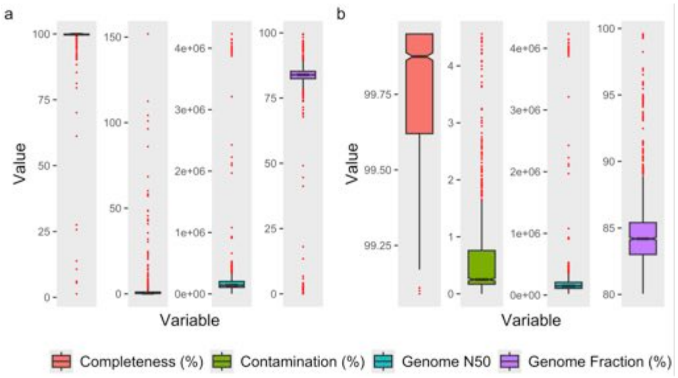


Figure. Quality of genome assemblies.

Population structure of the GWAS Dataset

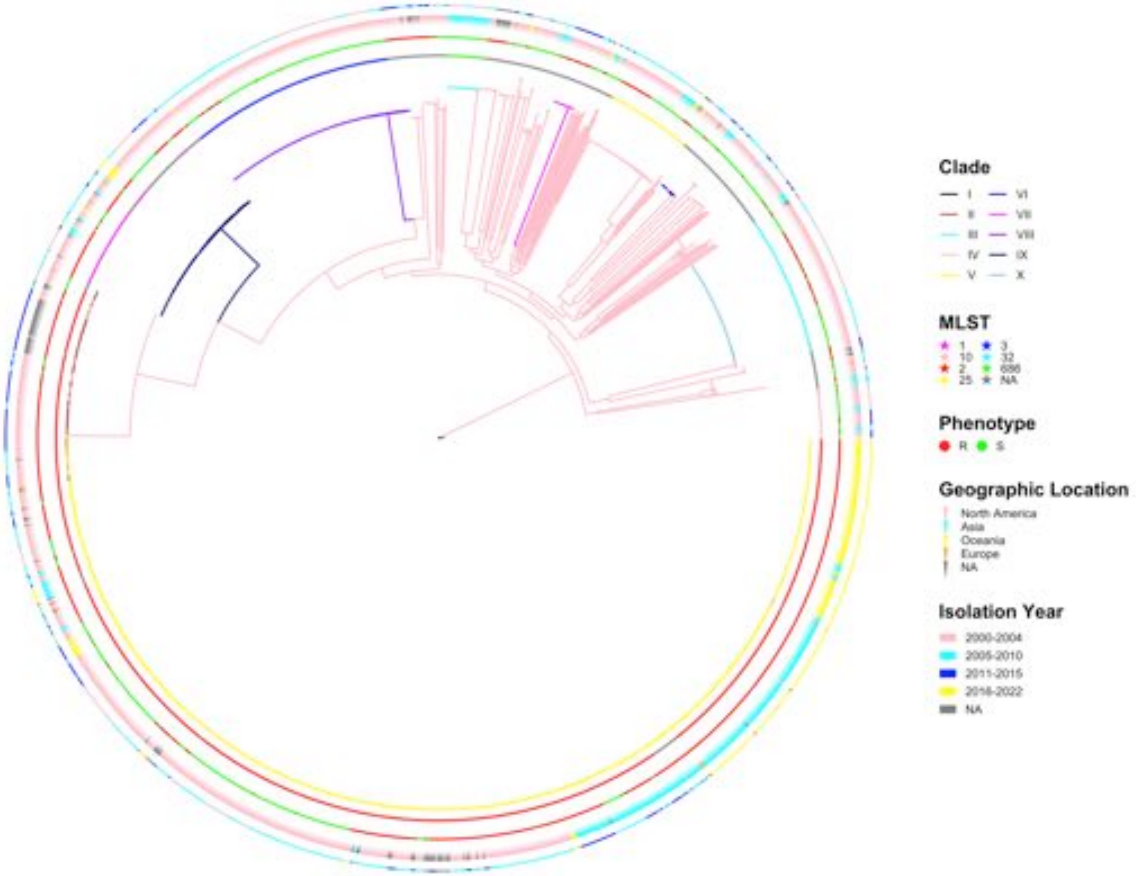


Figure. Core-genome-SNPs based phylogeny.

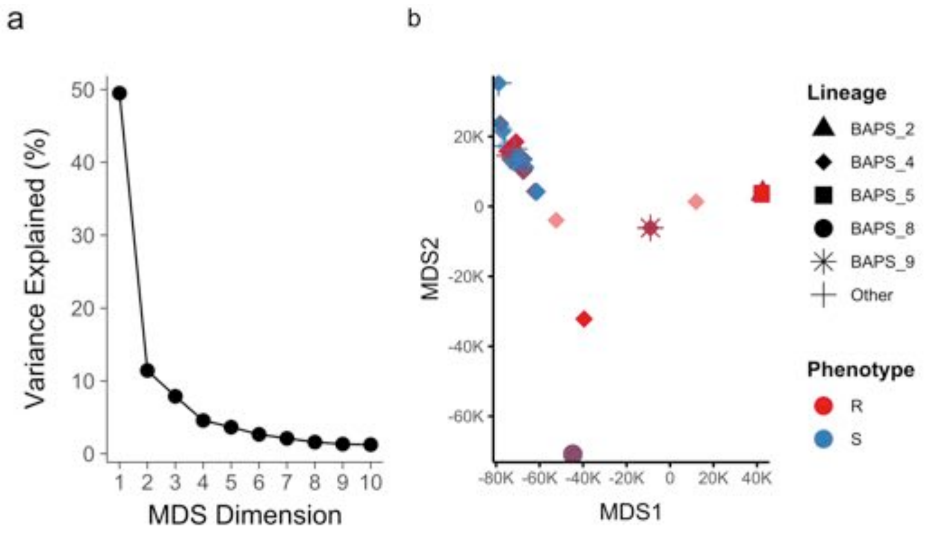
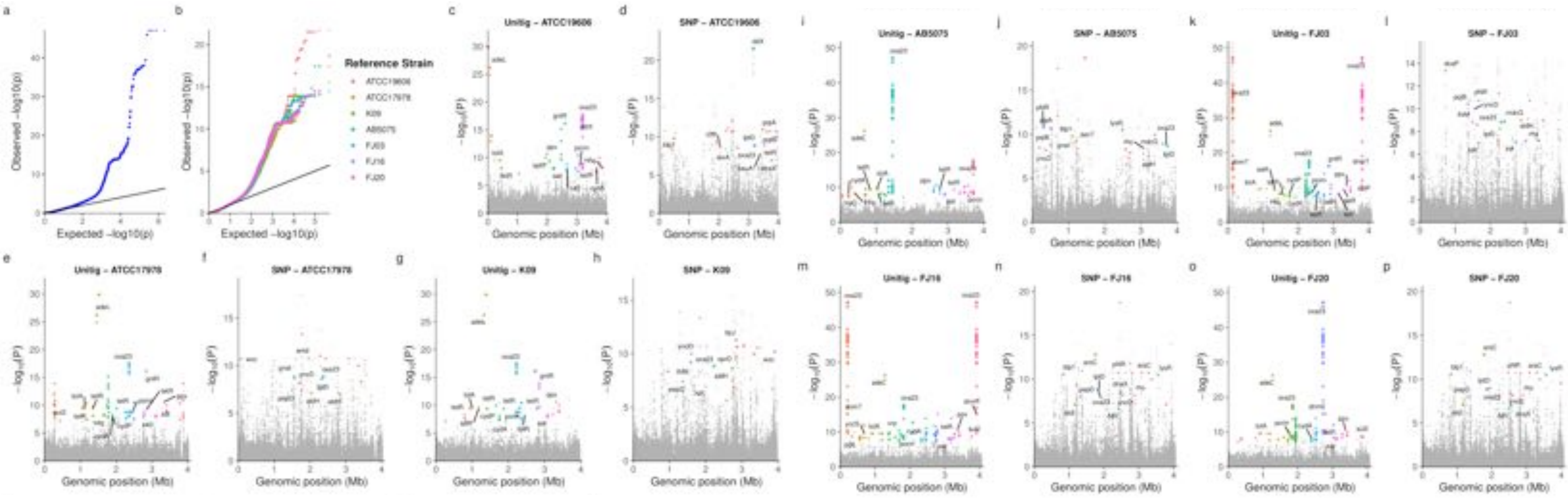


Figure. PCA of core-genome-SNPs.

Genetic *loci* associated with Carbapenem resistant



- Potentially novel resistance-associate mutations in outer-membrane porins (OprD, DcaP) and LPS-associated proteins (PqiB, LptD)
- Several resistance-associated mutations in hypothetical or poorly characterized proteins
- Expands the genetic landscape linked to carbapenem resistance in *A. baumannii*

The Identified Novel Mutations in Sequence & Structural Context

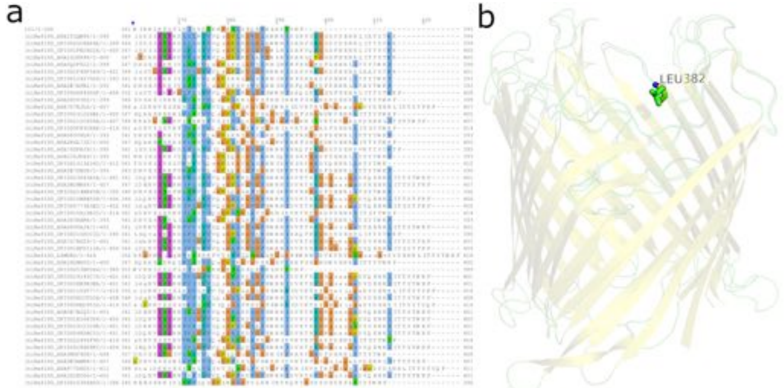
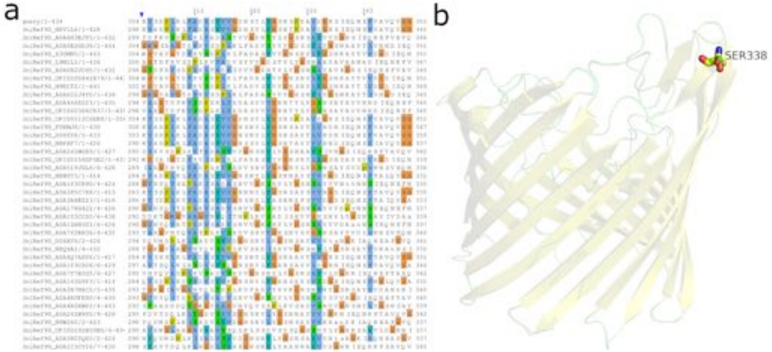


Figure. The Leu382Pro mutation identified in the outer-membrane porin, OprD



The ΔSer338 mutation identified in the outer-membrane porin, DcaP

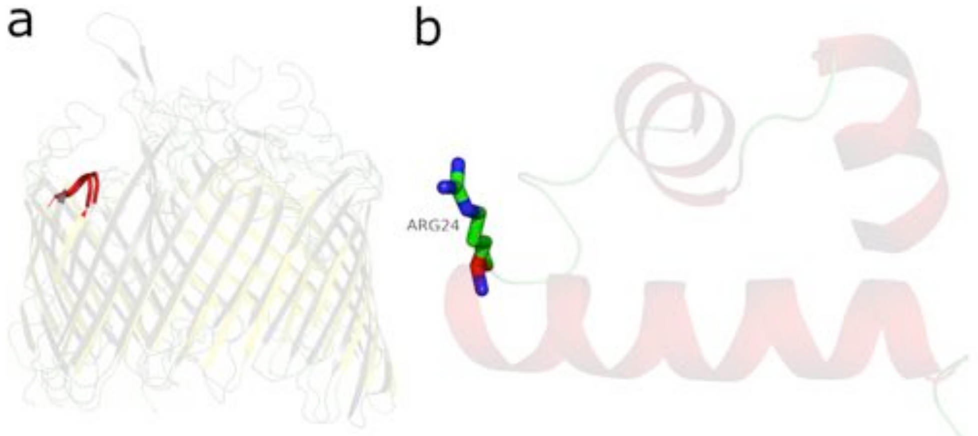


Figure. Mutations identified in LptD and TetR-family transcriptional regulator

- Limited sequence conservation
- Location of protein structure suggests plausible effect

Thank You